

AJAY KUMAR PONDUGALA

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BACKEND ENGINEERING | DISTRIBUTED SYSTEMS | CLOUD SECURITY

OBJECTIVE

Software Engineer II with 2+ years of engineering experience, including a six-month internship, building and validating Kubernetes-native backend capabilities for enterprise identity and network security. Hands-on with Go, Java, distributed services, observability, TLS/PKI, networking, incident diagnosis, and evidence-driven delivery, with verified scale testing to 500,000 endpoints. Strong in distributed systems, feature-level system design, RabbitMQ messaging, database-backed workflows, observability, and failure-mode analysis.

SKILLS

Backend & Design	Java, Spring Boot, Go, Python, REST APIs, microservices, distributed systems, feature-level system design, service boundaries and API/data contracts
Data & Messaging	RabbitMQ, asynchronous/event-driven processing, PostgreSQL, MySQL, MongoDB, OpenSearch, SQL, relational/document/search data modeling and mappings
Platform & Delivery	Kubernetes, Docker, Helm, Terraform, Linux, NGINX, GitLab CI/CD, Jenkins
Reliability & Security	OpenTelemetry, Prometheus, Grafana, Tempo, TLS/PKI, RADIUS, authentication/authorization, incident RCA
AI Engineering	Agentic CI triage, CVE-remediation prototypes, human-in-the-loop guardrails, prompt/context engineering, Claude Code, OpenAI Codex, Cursor, ChatGPT

EXPERIENCE

Software Engineer II Cisco	Sep 2025 - Present <i>Bengaluru, India</i>
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- **Kubernetes-native distributed security platform.** Build and validate Go and Java/Spring Boot service flows across enterprise authentication and authorization, TLS/PKI, RADIUS, RabbitMQ messaging, PostgreSQL persistence, and containerized dependencies.
- **Validated 500,000 endpoints in 350.8 seconds.** Confirmed queue drainage and exactly 500,000 persisted classifications in a Kubernetes scale run; both replicas remained Ready with zero restarts and no high-severity errors in the captured test window.
- **Validated restoration of access to 57 certificate keys.** Implemented and branch-validated a cross-service correction spanning a Java event producer, OpenSearch ingestion and mappings, Go contract tests, and a Rails API; eliminated all 14 observed empty attributes in a 22-attribute validation and passed 269 regression examples with zero failures.
- **Isolated a Kubernetes dependency failure.** Traced unavailable services to missing secret-generation resources that blocked RabbitMQ and downstream readiness; ruled out node, image-pull, DNS, and application causes and documented reusable recovery checks.
- **Qualified an OpenTelemetry Java-agent upgrade.** Validated an immutable candidate image end to end, confirming metrics, traces, service/version/resource attribution, and an mTLS dependency-health gauge transition; 506/506 service E2E tests and the dashboard E2E pipeline passed.
- **Feature-level system design and cross-service ownership.** Translated requirements into service, API, and data-contract changes plus implementation, validation, release, and recovery plans; coordinated fixes across distributed platform components.

Software Engineer Cisco	Aug 2024 - Sep 2025 <i>Bengaluru, India</i>
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- **Shipped enterprise certificate-lifecycle capabilities.** Developed Java/Spring Boot workflows for CA certificate management, validation, and stale-certificate detection in an identity-security platform.
- **Built database-backed service changes.** Implemented REST API and microservice changes across MySQL/MongoDB-backed workflows and data contracts, with Robot Framework/Python validation in CI/CD pipelines.

SELECTED AI ENGINEERING & INFRASTRUCTURE PROTOTYPES

Internal prototype work; branches were unmerged and not production-deployed.

- **Prototyped agentic CI and vulnerability triage.** Authored internal Python workflow branches that transformed GitLab pipeline and security-scan evidence into bounded, reviewable recommendations for CI failures and CVE remediation.
- **Engineered secure human-in-the-loop controls.** Implemented REST integrations with short-lived OIDC authentication, resumable Server-Sent Events, idempotent actions, sanitized evidence retrieval, and approval-gated Jira drafts.
- **Built fail-closed runtime guardrails.** Bound authorization, reviewed file scope, immutable image provenance, and per-advisory scan evidence; verified candidate images with isolated Docker Compose health, logs, metrics, and traces, backed by 23 contract and security tests.
- **Applied AI coding assistants with review guardrails.** Used Claude Code, OpenAI Codex, Cursor, and ChatGPT for specification-driven planning, codebase analysis, test debugging, refactoring, and PR review while retaining human validation for correctness and security.

EXPERIENCE - CONTINUED

Software Engineer

Cisco

Aug 2024 - Sep 2025

Bengaluru, India

- **Owned feature design and delivery end to end.** Converted requirements into feature-level designs, implementation and regression plans, release-readiness evidence, and cross-team defect resolution.
- **Improved regression confidence.** Investigated pipeline and system failures using logs and service behavior, separated product issues from test or environment issues, and verified fixes before merge.

Software Engineering Intern

Cisco

Jan 2024 - Jun 2024

Bengaluru, India

- **Built a QoS calculator and network-validation automation.** Automated policy mapping and network-latency analysis and contributed Robot Framework checks for IOS XE/XR and ACI environments.

ADDITIONAL RELEASE, SECURITY & SCALE EVIDENCE

- **Qualified an exact OpenTelemetry Collector upgrade.** Validated an isolated mTLS metric-and-trace pipeline; 454/454 exact-image qualification checks passed, with zero restarts and no High or Critical findings in the captured security scan.
- **Validated two immutable Go security-update images.** Each image passed 122/122 feature tests and 493/493 strict E2E checks; baseline comparison separated existing environment failures from candidate-image regressions.
- **Characterized a high-volume data boundary.** Designed a deterministic 12-scenario CSV suite spanning 10,000 to 99,997 rows, plus exact fixtures at 1, 74,896, 74,897, and 100,000 rows; isolated a 2 MiB Base64-string boundary and 20.6-30.8-second UI delays.

CERTIFICATIONS & EDUCATION

NVIDIA Certified Professional: AI Infrastructure - May 2026; valid through May 2028

NVIDIA Certified Associate: AI Infrastructure and Operations - Dec 2025; valid through Dec 2027

B.Tech, Computer Science Engineering - Gayatri Vidya Parishad College of Engineering, Visakhapatnam; CGPA: 9.3/10